Testing a Learner Changes It: Identifying Biased Self-Evaluation from Actions with Verifiable Probes

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Abstract

Language model agents increasingly learn from their own judgments of whether a task succeeded. If these judgments grow more lenient as the agent grows confident in an option, it can keep choosing a suboptimal option forever, and an auditor who sees only its choices cannot tell it from an unbiased agent facing a good option. We show that the auditor can detect this confirmation bias by sending verified probes, which are tasks whose outcome it knows, such as canary tasks with hidden tests. Watching alone leaves an error floor that more data does not lower, and probing at a fixed rate needs a number of rounds polynomial in $1/\delta$ for error level δ. We give the optimal audit. It sends verified successes, then verified failures, and reads the agent's choices once its belief crosses the value w of its default option. It needs $\log(1/\delta)/F(w)$ rounds, where F is an explicit rate. This matches a lower bound for every adaptive design and still holds when the auditor knows only a range for the agent's starting belief. With a known prior and $\delta = 10^{-2}$, the audit needs 16 rounds for a strong bias and 57 for a mild one, against 26 and 424 for random probes. It requires w to be known and every pull of the audited option to be a probe. Auditing self-grading agents therefore calls for canaries that look like real tasks and a known default value. For AI oversight, this shows that a self-grading agent can be audited from outside with a provable number of rounds, and that this is impossible from choices alone or without a known default value.

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1 INTRODUCTION

Language model agents increasingly learn from their own judgments. A coding agent with persistent memory decides whether its patch worked, stores the verdict, and relies on it later (Nous Research, 2026; Shinn et al., 2023), and self-rewarding training uses the model as its own grader (Yuan et al., 2024). Without ground truth, a skewed judgment shapes the next decision, and such skews are well documented. Models favor their own outputs (Panickssery et al., 2024; Xu et al., 2024), update more on positive than on negative outcomes (Schubert et al., 2024), and let their current belief predict how it changes (He et al., 2025), much as humans show confirmation bias (Palminteri et al., 2017; Lefebvre et al., 2017). *Can someone outside the agent detect this bias from its behavior, and at what cost?* We show that they can, with a provable and small number of interactions.

We study this in a bandit framework. An agent chooses repeatedly between an audited option, such as a code generator, and a default option of value w. It keeps a Beta belief about the audited option and chooses by Thompson sampling or a similar rule (Agrawal and Goyal, 2012; Kaufmann et al., 2012; Russo et al., 2018). It grades each outcome itself, and it is confirmation-biased if it reads a failure as a success more often when its belief is higher. Such an agent can lock in on a suboptimal option, because its grades keep the belief high (Section 4).

Prior work infers rewards from learning agents and treats their suboptimality as transient (Jacq et al., 2019; Ramponi et al., 2020), which a persistent bias is not. The outside party, the auditor, sees only the agent's choices. It can also submit probes, tasks whose outcome it knows, such as canary tasks with hidden tests, a known-answer device from crowdsourcing (Hettiachchi et al., 2021). Sequential lower bounds use a change of measure (Kaufmann et al., 2016; Garivier et al., 2019) and assume observations that ignore the design (Naghshvar and Javidi, 2013). Here each probe changes the agent's belief and so its later choices.

Our main result is the optimal audit. A test needs at least $\log(1/\delta)/F(w)$ rounds for error level δ, where $F(w)$ is an explicit rate set by how often the agent reads a failure as a success and by w (Theorem 5). A fill-drain design attains it. It sends verified successes, then verified failures, because a failure sent while the agent is confident carries the most information. It reads the agent's choices once an unbiased agent would have stopped pulling the option while a biased agent still pulls it (Theorem 6). It stays worst-case optimal when the auditor knows only a range for the agent's prior (Theorem 7). The auditor must know w. Without it, no design and no test can push the error below a fixed constant, however long it runs (Proposition 3). Every pull of the audited option must be a probe.

Two simpler audits fail. Watching the choices leaves an error floor that more data does not lower (Proposition 2), and probing at a fixed rate costs polynomially many rounds in $1/\delta$, because probes change the agent's belief (Theorem 4). In simulation the audit needs far fewer rounds than random canaries (Section 7). It fails when only some outcomes are canaries, and it flags honest agents when their noise is not budgeted.

For the oversight of agents that learn from self-evaluation, this gives a guarantee on both sides. An external audit is possible with a provable number of rounds, and it is impossible from choices alone or without a known default value. Teams that build or oversee such agents can run a short audit, provided the canaries look like the agent's own tasks, the default value is fixed by the audit environment, and a noise budget covers honest errors.

Contributions.

- A model of belief-dependent self-evaluation bias that abstracts biases already measured in LLMs, and an auditor that sees choices and verified probes only (Section 3).
- Passive tests have an error floor, no test works without a known w, and fixed-rate probing needs polynomially many rounds (Propositions 2 and 3, Theorem 4).
- An optimal audit. A $\log(1/\delta)/F(w)$ lower bound for every design, adaptive or not, and a fill-drain design that attains it, worst-case optimal for an unknown prior, with a noise budget for honest errors (Theorems 5–7, Proposition 8).
- Simulations of the audit and a stress test of each assumption (Section 7).

2 RELATED WORK

Biases of LLM self-evaluation. The existence of these biases in language models is well established, and our model takes them as its starting point. Language models prefer their own outputs when they act as judges (Panickssery et al., 2024), and self-refinement increases this bias (Xu et al., 2024). Self-rewarding schemes rely on the model as its own grader (Yuan et al., 2024), and self-training with self-generated rewards can collapse (Shafayat et al., 2025). Schubert et al. (2024) measure asymmetric belief updating in in-context learners, and He et al. (2025) measure belief entrenchment with the Martingale Score. These works measure a bias with access to the model, to its probabilities, or to ground truth. We ask how much of it an outside party can identify from choices, and at what cost.

Confirmation bias in learning models. Human reinforcement learners update more after confirming than after disconfirming outcomes (Palminteri et al., 2017; Lefebvre et al., 2017; Palminteri and Lebreton, 2022). The standard model uses two learning rates. Ours acts on the perceived outcome and depends on the current belief, which is what makes the bias persistent in a bandit.

Learning from learners. Inverse reinforcement learning from a learning agent (Jacq et al., 2019; Ramponi et al., 2020) treats suboptimality as a transient effect that fades with data. A known irrationality model can help or hurt reward inference (Chan et al., 2021), and some models cannot be separated from behavior (Skalse et al., 2023). Our bias does not fade, and we study when it can be identified at all.

Sequential testing and active design. Our lower bounds use the change of measure of Kaufmann et al. (2016) and Garivier et al. (2019). Active sequential testing and controlled sensing give optimal designs when observations are independent given the action (Naghshvar and Javidi, 2013; Nitinawarat and Veeravalli, 2015). Here the agent adapts to the probes, so the observations depend on the whole history, and the rate $F(w)$ differs from a plain information rate. Anytime-valid inference (Ramdas et al., 2023) gives type-I control for tests of this kind.

Bandit learners. Thompson sampling (Agrawal and Goyal, 2012; Kaufmann et al., 2012) and count-based confidence bounds (Auer et al., 2002) are the learners we audit. Their regret is well understood for unbiased feedback (Lai and Robbins, 1985). Under confirmation bias they can lock in on a suboptimal option, which we use in Section 4.

3 SETUP

Agent. At each round t the agent chooses between an audited option ($a_t = 1$) and a default option ($a_t = 0$) of known value $w \in (0, 1)$. The audited option has an unknown success rate. The agent holds a belief $\mathrm{Beta}(\alpha_t, \beta_t)$ with prior $(\alpha_1, \beta_1) = (A, B)$ and posterior mean $m_t = \alpha_t/(\alpha_t + \beta_t)$. After a pull it records a perceived outcome $\tilde{y}_t \in \{0, 1\}$ and updates $\alpha_{t+1} = \alpha_t + a_t\tilde{y}_t$ and $\beta_{t+1} = \beta_t + a_t(1 - \tilde{y}_t)$. The choice rule is sharp at w. It pulls the audited option with probability close to one when m_t is above w and close to zero when it is below. Thompson sampling ($a_t = \mathbf{1}\{X_t > w\}$ with $X_t \sim \mathrm{Beta}(\alpha_t, \beta_t)$), greedy choice, and count-based confidence bounds are sharp in this sense.

Probes. Before round t the auditor picks a probe rate $q_t \in [0, 1]$ that may depend on everything it has seen. If $a_t = 1$, the outcome the agent grades is a verified probe $x_t \sim \mathrm{Bern}(q_t)$ whose value the auditor knows. If $a_t = 0$ nothing is graded. In the coding example, a probe is a canary task with hidden tests. The auditor observes (a_t, x_t) and never $\tilde{y}_t$. It is charged for rounds τ and for probes $N_\tau = \sum_{t\le\tau} a_t$.

Two agent types. The unbiased agent M_0 records $\tilde{y}_t = x_t$. The biased agent M_1 is a confirmation-biased agent with a kernel K. It records a success as a success, $K(1, m) = 1$, and it records a failure as a success with probability $K(0, m) = g(m)$ when its belief is m.

Assumption 1 (Kernel)**.** A success is never misread, $K(1, m) = 1$. The leniency g is nondecreasing in m and stays below 1 on $[0, 1)$. For the audit of Section 6, the auditor only needs a lower bound $g_{\min} \le g$ on the leniency, which sets the weakest bias it wants to detect.

The canonical example is $g(m) = \min\{1, \max(\gamma m,\, 1 - C(1 - m)^p)\}$ with (γ, C, p) set to $(.5, 1, 2)$ for a strong bias, $(.3, 20, 2)$ for a mild bias, and $(.1, 5, 2)$ for a weak bias. We call these the strong, mild, and weak kernels. A leniency that does not depend on belief is the same as an honest grading noise, and we treat it as part of the null in Section 6.

Evidence per probe. Take a stationary probe rate q and an agent whose belief equals q. Each probe then carries the evidence $e_K(q) = q \log \frac{1}{K(1,q)} + (1 - q)\,\ell_0(q)$ with $\ell_0(m) = \log \frac{1}{1-g(m)}$. This is the Kullback-Leibler divergence between the perceived outcomes of the unbiased and the biased agent. We define the rate

$$F(m) \;=\; m \int_m^1 \frac{\ell_0(x)}{x^2}\, dx \;\ge\; e_K(m). \qquad (1)$$

For the strong kernel at $w = \frac{1}{2}$ we have $F(w) = 1.386$ and, in the plain evidence units, $e_K(w) = \log 2$.

Tests. A test is a stopping time τ with a decision $\hat{z} \in \{0, 1\}$ based on $(a_t, x_t)_{t\le\tau}$. It is δ-correct if $\mathbb{P}_{M_z}(\tau < \infty) = 1$ and $\mathbb{P}_{M_z}(\hat{z} \ne z) \le \delta$ for $z \in \{0, 1\}$. We write $b = \log(1/\delta)$ and report the expected number of rounds. The null is perfect self-evaluation. An agent with a noisy grader whose noise does not depend on belief is a different null that we treat in Section 6.

Assumptions on the audit. The following assumptions hold throughout, and Section 7 tests each of them.

(H1) Two options with binary verified outcomes, and a Beta belief updated by counts.

(H2) The choice rule is sharp at w. At a mean below (above) w by a fixed margin, it pulls with a probability that is exponentially small (close to one) in the number of pulls.

(H3) The auditor knows w. Proposition 3 shows this cannot be dropped.

(H4) The prior lies in a known box, and Assumption 1 holds.

(H5) Every pull of the audited option is a probe, and the agent cannot tell a canary from a real task.

(H6) The auditor cannot read, copy, or reset the agent's state.

Audit access. The agent's grades are private, and they are the quantity that may be biased, so the auditor cannot use them. It cannot read, copy, or reset the agent's state, as for a deployed model with its own memory. It controls the verified outcomes the agent experiences, and it sees whether the agent keeps choosing the option. A canary that the agent can recognize would be graded correctly and tell nothing, so (H5) constrains the auditor's tasks. The default value w is the agent's internal indifference level. It is known when the audit environment fixes and announces it, or when a control agent of the same model, fed true verdicts, can calibrate it.

4 WATCHING IS NOT ENOUGH

Suppose the auditor sends no probes and only watches the choices. The audited option has success rate $\mu < w$, so an unbiased agent should stop pulling it. A confirmation-biased agent may not. Its early failures are partly read as successes, its belief stays above w, and it keeps pulling a suboptimal option.

Proposition 2 (Confound)**.** *Let g have a lock-in level, that is, $g > 0$ and $g(m) \ge m$ on $[\ell, 1)$ for some $\ell \in (w, 1)$. This holds for the canonical kernels. There is $\rho > 0$ such that for every horizon T the biased agent*

facing the option of rate μ and the unbiased agent facing an option of rate 1 both pull the audited option at all T rounds with probability at least ρ. Every test that uses only the choices, including sequential tests, has error sum at least ρ. No such test is δ-correct for $\delta < \rho/2$.

For the strong kernel at $\mu = 0.3$, $w = \frac{1}{2}$ and prior $(2, 2)$ we get a floor of 0.0943. For the mild kernel it is 1.99×10^{-6}. The error sum is the sum of the two error probabilities, so no test is δ-correct below $\rho/2$. For the mild kernel the floor matters only at very small δ, but more observation does not lower it.

The proof compares the two probabilities of the all-pull path. Under the biased agent, the belief drifts to a lock-in level above w with positive probability. Under the unbiased agent facing a perfect option, the belief rises at every pull. Both paths exist at every horizon, and the probabilities do not vanish with T.

The default value cannot be inferred. The auditor needs to know the threshold at which the agent switches. Suppose it only knows that the threshold lies in $(0, 1)$.

Proposition 3 (No anchor)**.** *Let g have a lock-in level. If the auditor does not know the agent's default value w, every design and every test has error sum at least $p_0 > 0$.*

For the strong kernel at $w = \frac{1}{2}$ with prior $(2, 2)$ we get $p_0 = 0.12$. The reason is that a strongly confirming agent with threshold w and an unbiased agent with a different threshold put the same mass on the same paths. Choices identify only a combination of the threshold and the leniency, so the two cannot be separated. This is why (H3) is an assumption and why the audit needs an anchor. The auditor can fix and announce the default, or use a control agent that is fed true verdicts.

5 TESTING CHANGES THE LEARNER

When the auditor sends probes, it changes what the agent believes, and the agent changes what it chooses. This section shows what happens with the simplest design and then gives the optimal one.

5.1 Stationary probing is slow

A stationary design sends probes at a fixed rate q. Under it, the unbiased agent's belief converges to q. If $q < w$, it stops pulling the option, and the audit has few probes to read.

Theorem 4 (Stationary probes)**.** *Fix $q \in (0, 1)$ and let K satisfy conditions (K1), (K2), and (K5) of Appendix A. Every δ-correct test satisfies*

$$\mathbb{E}_{M_0}[N_\tau] \;\geq\; (1 - o(1))\, \frac{\log(1/\delta)}{e_K(q)}.$$

If $q < w$ it also satisfies $\mathbb{E}_{M_0}[\tau] \geq \delta^{-\theta_K(q)+o(1)}$ with $\theta_K(q) = \mathrm{kl}(q, w)/e_K(q)$, where kl is the Bernoulli Kullback-Leibler divergence. At $q = w$ it satisfies $\mathbb{E}_{M_0}[\tau] \geq c \log^2(1/\delta)$ for a constant $c > 0$. For one-sided kernels and $q < w$, the exact sequential likelihood ratio test attains the probe bound and the round bound.

The number of probes grows like $\log(1/\delta)$ at any rate. The number of rounds does not, because the agent skips the option and each skipped round returns no probe. A stationary rate below w gives a polynomial number of rounds in $1/\delta$.

5.2 The optimal audit

The auditor can do better by changing q_t over time. The idea is to make the agent confident first, then send failures while it is still confident, and read its choices once its belief reaches w.

Fill-drain design. Fix r. The first r pulls receive verified successes ($x = 1$), and every later pull receives a verified failure ($x = 0$). The unbiased agent, whose mean after r successes is above w, then slides below w at a pull index n_c that the auditor computes from the prior. The window test watches the W rounds after the n_c-th pull, with failures still being sent, and declares the agent biased if it pulls the option at least M times. For the strong kernel, a known prior $(2, 2)$ and $\delta = 10^{-2}$, the calibrated design sends 4 verified successes, watches the 8 rounds after the 6-th pull, and flags the agent if it pulls at least 7 times. For the mild kernel the four numbers are 23, 13, 41, and 11.

An unbiased agent stops pulling soon after n_c. A biased agent has read some of the early failures as successes, so its belief is higher, and it keeps pulling. The failures sent while the belief is high carry the most evidence, which is why the successes come first.

Theorem 5 (Lower bound)**.** *Under (H1) to (H5), for any prior, every adaptive and randomized design and every δ-correct test satisfy*

$$\mathbb{E}_{M_0}[\tau] \;\geq\; (1 - o(1))\, \frac{\log(1/\delta)}{F(w)}.$$

Theorem 6 (Fill-drain is optimal)**.** *Under (H1) to (H5) with a known prior and a Thompson sampling agent, there are parameters (r, W, M) depending on δ*

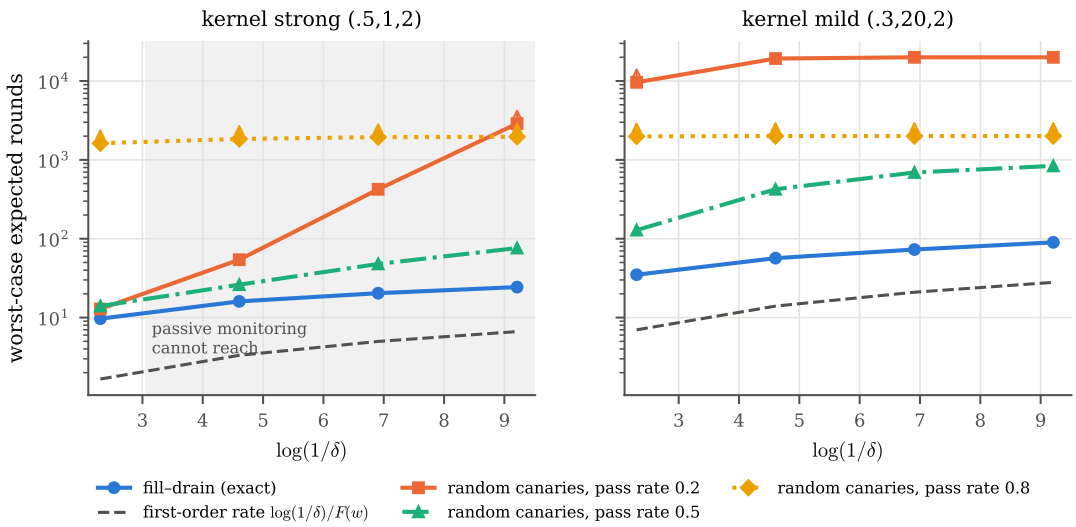


Figure 1: Expected number of rounds to reach error δ for the strong and mild kernels. Fill-drain is computed exactly. Random canaries use pass rates 0.2, 0.5, and 0.8 and 100 Monte Carlo paths per setting, capped at 20000 rounds and 2000 probes. The passive floor ρ from Proposition 2 is the error level that watching alone cannot cross.

such that the fill-drain design with the window test is δ-correct and, for $z \in \{0, 1\}$,

$$\mathbb{E}_{M_z}[\tau] \;=\; \frac{\log(1/\delta)}{F(w)} + O\Big(b^{2/3} \log b\Big)\,.$$

For greedy and count-based learners, the same design makes no false alarm and is optimal among deterministic designs.

The rate in Theorem 5 does not depend on the prior, and Theorem 6 matches it to first order.

The window works for the following reason. The design sends enough failures before the crossing that their total evidence exceeds b by a margin of order $b^{2/3} \log b$. Under M_0 the belief is then just below w at the window, so the agent pulls fewer than M times with probability at least $1 - \delta$. Under M_1 the agent looks like M_0 only if its grader misreads very few of these failures, and the evidence collected makes this event as unlikely as δ.

The lower bound has three ingredients. A change of measure shows that the auditor needs total evidence b in the choices. A failure sent at belief m yields at most $\ell_0(m)$ units of evidence, and honest belief depends only on counts. Among all orderings of successes and failures, the sorted one, successes first, keeps the belief highest for the longest, and an integral over the belief path gives the rate $F(w)$. Appendix A gives the details.

To see where $F(w)$ comes from, take r successes followed by failures. After j pulls the honest mean is $R/(s + j)$ with $R = A + r$, and a failure at mean m carries $\ell_0(m)$ nats. Summing ℓ_0 along this path and substituting $x = R/(s + j)$ gives $R \int \ell_0(x) x^{-2} dx$. At the crossing $R \approx w(s + n)$, so the sum is about $nF(w)$ after n pulls. The rate $F(w)$ is the evidence per pull that the belief collects while it slides down to w.

6 AN AUDIT THAT WORKS WITHOUT THE AGENT'S PRIOR

The fill-drain design of Theorem 6 uses the agent's prior to locate n_c. A deployed auditor does not know it. We assume only that the prior lies in a known box $[A_{\text{lo}}, A_{\text{hi}}] \times [B_{\text{lo}}, B_{\text{hi}}]$ and that the leniency is at least a known floor $g_{\min}$. Let $F_{\min}$ be the rate F of Equation (1) computed for $g_{\min}$. The auditor must be δ-correct for every prior in the box and every kernel above the floor.

Theorem 7 (Minimax audit)**.** *Let the auditor know w, the box, and $g_{\min}$. A design and test are uniformly δ-correct if both error probabilities are at most δ for every prior in the box and every kernel with $g \geq g_{\min}$. The minimax number of rounds over such designs and tests, with the supremum over the same priors and kernels, satisfies*

$$\inf \sup\; \mathbb{E}[\tau] \;=\; (1 + o(1))\, \frac{\log(1/\delta)}{F_{\min}(w)}.$$

The infimum is attained by a fill-drain design whose window is anchored at the pessimistic prior and whose threshold covers the optimistic prior.

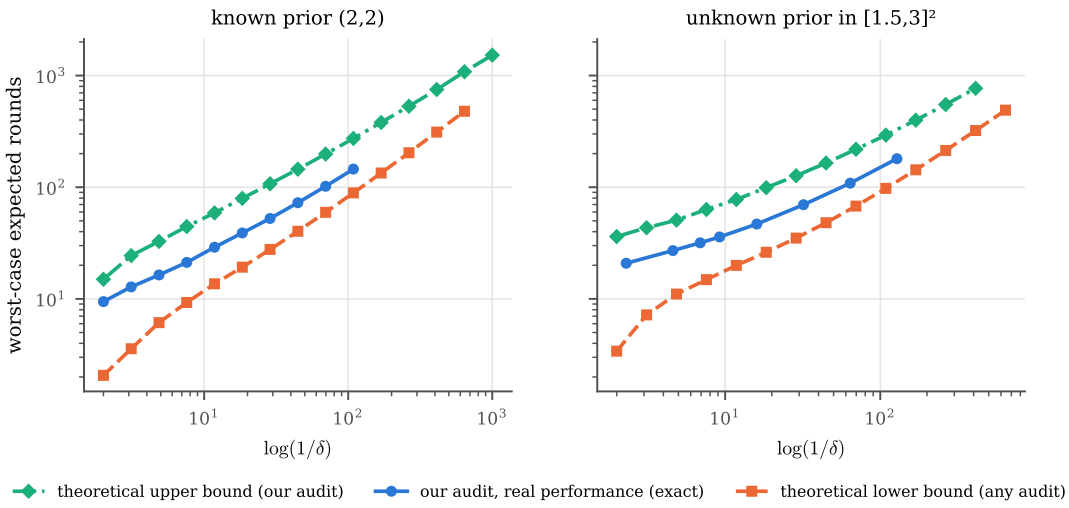


Figure 2: Rounds of the calibrated audit (blue), a lower bound valid for every δ (orange), and an upper bound (green), against $\log(1/\delta)$ for the strong kernel. Left, known prior $(2, 2)$. Right, prior in the box $[1.5, 3]^2$.

The upper bound uses a coupling. A higher prior mean keeps the honest agent pulling longer, so the false-alarm rate of the window test is largest at the optimistic corner of the box. A lower leniency keeps the biased belief lower, so the miss rate is largest at the pessimistic corner with kernel $g_{\min}$. Calibrating the window for these two cases covers every prior in the box and every kernel above the floor. The price of the unknown prior is a number of rounds that does not grow with δ. The two corners of the box are the worst cases for the two kinds of error, so calibrating there covers the whole box. In practice we calibrate by exact dynamic programming over these corners. The resulting audit is valid at every δ, not only in the limit. At $\delta = 10^{-2}$ and the box $[1.5, 3]^2$ it takes 27 rounds for the strong kernel, against 23 when the least favorable prior is known. For the mild kernel the numbers are 81 and 72. The first-order value $b/F(w)$ is 3.3 and 14.0, so first-order constants understate the practical cost by a large factor. The gap comes from the lower-order terms of the upper bound, which dominate at moderate δ.

Honest noise. A grader can be wrong without being biased. A leniency that does not depend on belief is the same as an honest noise that flips failures into successes, and no test on choices can separate the two. We therefore state the null with a noise budget.

Proposition 8 (Noise budget)**.** *Suppose an honest agent reads a failure as a success with probability $v \le v_{\max}$, independent of its belief. If the fill-drain test is calibrated for an honest agent with noise $v_{\max}$, its false-alarm probability is at most δ for every $v \le v_{\max}$.*

The auditor pays for this safety with a larger design. Table 1(b) gives the cost at $\delta = 10^{-2}$. A noise budget of 0.2 raises the rounds from 16 to 23 for the strong kernel and from 57 to 118 for the mild one. The audit detects a bias only if the leniency exceeds the noise budget at the beliefs the agent visits.

Where the guarantees stop. Theorem 5 covers choice rules that are sharp at w. A softmax rule with a fixed temperature is outside it, and Section 7 tests how far the audit tolerates one. Without a known w the audit has no meaning (Proposition 3). If the agent forgets old evidence, the belief at the window is lower than the audit expects. If only some of the agent's outcomes are probes, the agent's belief is partly driven by experience the auditor cannot control. Section 7 measures the effect of each.

7 EXPERIMENTS

The experiments simulate the model of Section 3. They check the bounds, compare the audit with common practice, and break each assumption in turn. We use the strong, mild, and weak kernels, $w = \frac{1}{2}$, and Thompson sampling agents. Quantities that the theory can compute exactly, such as the error of a design for a given prior, come from exact dynamic programming. The others come from simulation.

7.1 Testing changes the learner

We compare the fill-drain audit with a random-canary audit. The random canary sends a probe with a fixed pass rate q and applies an exact sequential likelihood

pass rate q	strong	mild
0.2	54	19292*
0.3	38	2515*
0.4	31	984*
0.5	26	424
0.55	62	208
0.6	305*	934*
0.8	1843*	2006*
fill-drain	16	57

(a) Fixed pass rates and fill-drain.

noise budget $v_{\max}$	strong	mild
0	16	57
0.05	18	69
0.1	19	81
0.2	23	118

(b) Fill-drain with a noise budget.

Table 1: Worst-case expected rounds at $\delta = 10^{-2}$ for a known prior $(2, 2)$. In (a), a star marks rates where more than 5% of the runs reach the cap, so the value is a lower bound. The cap is 20000 rounds for $q \in \{0.2, 0.5, 0.8\}$ and 3000 rounds for the other rates. In (b), the audit is calibrated for honest graders that read a failure as a success with probability up to $v_{\max}$, and it never reads a success as a failure.

ratio test with the same error level. Figure 1 shows the worst case over the biased and unbiased agent, for a known prior $(2, 2)$.

At $\delta = 10^{-2}$ the fill-drain audit needs 16 rounds for the strong kernel and 57 for the mild one. Random canaries at pass rate 0.5 need 26 and 424. A pass rate of 0.2 gives 54 rounds for the strong kernel and fails to finish for the mild one, where a fraction 0.94 of the runs is still running at the cap. This is the effect of Theorem 4 for a rate below w. At 0.8 the agent keeps pulling, but failures are rare and each probe carries little evidence. A fraction 0.92 of the strong-kernel runs and 1.00 of the mild-kernel runs remain unfinished at the cap. Table 1(a) sweeps the pass rate. The best fixed rate is 0.5 for the strong kernel, with 26 rounds, and 0.55 for the mild kernel, with 208 rounds. The fill-drain audit needs 1.6 times fewer rounds on the strong kernel and 3.7 times fewer on the mild one. The passive floor is 0.0943 for the strong kernel, and a passive test cannot reach any smaller δ.

7.2 An auditor with an unknown prior

Figure 2 compares the real cost of the minimax audit with the theoretical bounds for all values of $\log(1/\delta)$. The audit is calibrated by exact dynamic programming at the corners of the prior box. The curves are the finite-δ lower and upper bounds of Appendix A.3. The audit stays between them at every value we computed.

The unknown prior adds a constant number of rounds. At $\delta = 10^{-2}$ the box audit takes 27 rounds for the strong kernel, 81 for the mild one, and 42 for the weak one, against 23, 72, and 39 for an auditor that knows the worst prior. The box $[1, 4]^2$ costs 49, 116, and 70. Appendix B shows that the ratio of the audit cost to $b/F_{\min}$ approaches one slowly. It is 1.95 at $b = 128$.

7.3 Stress test

We now break the assumptions one at a time at $\delta = 10^{-2}$. Each point in Figure 3 uses 20000 runs of each agent type on the box audit. A missed detection is a biased agent that the test declares unbiased. A false alarm is the opposite. A value of 10^{-5} in the plot means that no error occurred in the 20000 runs.

On the strong kernel the audit gives 0.00015 false alarms and 0.00075 missed detections at the nominal setting. Both error rates stay below δ for softmax choice at temperature 0.1 (0.00235 false alarms, 0.0059 missed detections) and for an agent threshold above w. An agent threshold $w' = 0.45$ below w gives 0.00945 false alarms, and a safety margin on the window removes them. Appendix B varies the offset of w' and the level δ. Three departures push an error rate above δ. Honest grading noise at rate 0.1 gives 0.0465 false alarms, which is the case that Proposition 8 budgets for. A prior outside the box gives 0.0544 missed detections. Forgetting with a discount of 0.95 gives 0.0029 missed detections on the strong kernel and 0.979 on the mild kernel. Partial probing is the most damaging departure. When only 80% of the outcomes are canaries, the false-alarm rate is 0.123 for the strong kernel and 0.835 for the mild kernel. The audit needs (H5), and the other assumptions leave more room.

8 DISCUSSION

We studied how an outside auditor can detect an agent that grades itself more leniently as its belief grows, when it sees choices and verified probes only. The optimal audit costs $\log(1/\delta)/F(w)$ rounds and keeps this rate when the prior is unknown, up to a constant that does not grow with δ.

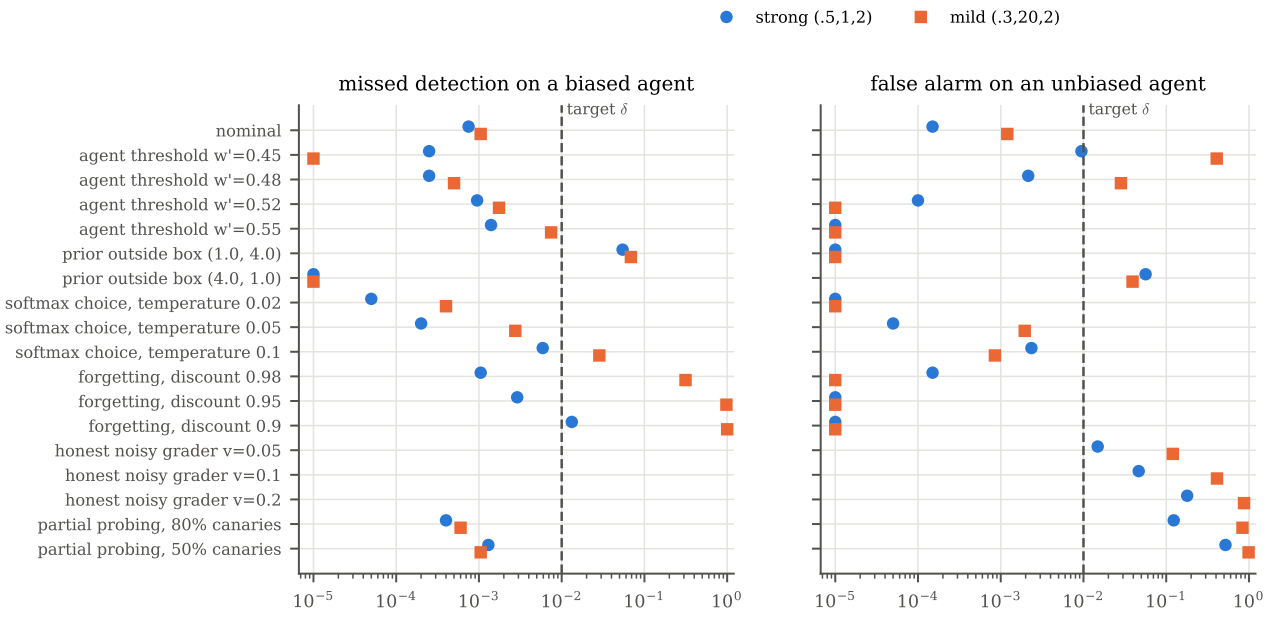


Figure 3: Missed detections on the biased agent (left) and false alarms on the unbiased agent (right) when one assumption is broken. Each row is a departure from the model. Values at 10^{-5} mean that no error occurred in 20,000 runs. The nominal error level is $\delta = 10^{-2}$.

What the audit needs from a deployment. The auditor must be able to send tasks with known outcomes that the agent treats as its own, and it must know the agent's default value. Both can be arranged by design. The audit environment can fix the default, and a control agent can calibrate it. The tasks must be indistinguishable from real ones, for example by drawing them from the same distribution as the agent's own tasks and grading them in advance. A canary that the agent can recognize tells nothing, and one that only some outcomes follow makes the audit fail, as Figure 3 shows.

Scope of the theory. The model has two options, a Beta belief, and a choice rule that is sharp at w. Fixed-temperature softmax and ε-greedy rules are outside Theorem 5. The bias is a belief-dependent leniency toward failures. We do not cover other distortions, such as a lower weight on negative evidence, a different reading of ambiguous outcomes, or a bias that depends on time. A flat leniency cannot be told from honest grading noise, so the auditor budgets for the noise it is willing to ignore, which costs the extra rounds of Table 1(b).

Beyond the guarantees. The finite-δ bounds of Appendix A.3 keep the audit between a lower and an upper curve at every δ we computed. Closed-form constants in the upper bound are only useful at extremely small δ, so practical numbers come from exact calibration.

Use with LLM agents. The biases that motivate the model, self-preference (Panickssery et al., 2024), belief entrenchment (He et al., 2025), and asymmetric updating (Schubert et al., 2024), are already measured in LLMs, so a kernel for a given model can be estimated from graded canaries. The theory then predicts the cost of the audit through $F_{\min}(w)$. It also predicts that the leniency of a self-grading agent rises with its belief, since a flat leniency is the honest noise of Proposition 8, and that choices near w are sharp at temperature 0. Each of these can be checked on a deployed agent, and a mismatch would show which assumption of Section 3 fails for that agent.

Future work. Extensions include more than two options, softmax and ε-greedy rules, estimating w from a control agent, and other distortions such as a lower weight on negative evidence.

AI use statement

We used generative AI tools to help draft and edit the text, to write and run simulation code, and to check derivations and formal proofs. We reviewed all AI-assisted work. The authors checked the simulation code against exact computations, and the proofs were checked line by line. We take responsibility for the final content of this work.

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Testing a Learner Changes It: Supplementary Materials

A THEORY DETAILS

This appendix gives the regularity conditions, a sketch of the proof of each result of the main text, and the finite-δ bounds that the main text refers to. The sketches give the main steps and the key estimates. The complete proofs are in the full version of the paper.

A.1 Notation and conditions

We write $\mathrm{kl}(p, q)$ for the Bernoulli Kullback-Leibler divergence and $g(m) = K(0, m)$, so that $\ell_0(m) = \log \frac{1}{1-g(m)}$. The unbiased honest belief after j pulls with s successes depends only on (j, s). Let μ_j be its mean. All proofs use the following conditions on the kernel, together with $K(1, \cdot) \equiv 1$.

(K1) g is nondecreasing and $g < 1$ on $[0, 1)$.

(K2) There is $c_K < \infty$ with $\ell_0(m) \le c_K\big(1 + \log \frac{1}{1-m}\big)$ on $[0, 1)$.

(K3) $F(w) > 0$.

(K4) (Lock-in level.) $g > 0$ on $(0, 1)$, and there is $\ell \in (w, 1)$ with $g(m) \ge m$ for every $m \in [\ell, 1)$.

(K5) At the probe rate q under study, g is continuous at q and $0 < g(q) < 1$, so that $e_K(q) > 0$.

Condition (K1) is the form of Assumption 1 that we use. The rate $F(w)$ integrates ℓ_0 on $[w, 1)$, so $g < 1$ is needed there. The proofs of Propositions 2 and 3 use (K4), and the proof of Theorem 4 uses (K5). The proofs of Theorems 5 and 6 use (K1) to (K3). For Theorem 7, (K1) to (K3) hold for the floor $g_{\min}$ in place of g, with $\ell_{\min}$ and $F_{\min}$. The canonical kernels satisfy (K1), (K2), (K4), and (K5) at every q with $g(q) > 0$, and (K3) for the three parameter sets of Section 3.

Throughout, $b = \log(1/\delta)$, $s = A + B$ for the prior (A, B), and $G(\alpha, \beta) = \mathbb{P}(\mathrm{Beta}(\alpha, \beta) > w)$ is the pull probability of a Thompson sampling agent in state (α, β).

A.2 Proof sketches

Common tools. Four facts are used repeatedly.

(T1) *Beta facts.* $G(\alpha, \beta)$ is nondecreasing in α and nonincreasing in β (Gamma representation of the Beta law). For integers, $G(\alpha, \beta) = \mathbb{P}(\mathrm{Bin}(\alpha+\beta-1, w) \le \alpha-1)$. With Hoeffding's inequality this gives exponential tails: if the mean is at most $w - \eta$ the pull probability is at most $e^{-(N-2)\eta^2/2}$ with $N = \alpha + \beta$, and symmetrically for a skip when the mean is at least $w + \eta$. So Thompson sampling is *sharp* at w.

(T2) *Honest cost and change of measure.* For a history h_t, let the honest cost $C_t(h_t)$ be the sum of $\ell_0(\mu)$ over the pulls that received a failure probe, with μ the honest belief before that pull. Keeping only the term of M_1 in which every probe is perceived correctly gives $\mathbb{P}_{M_1}(h_t) \ge e^{-C_t(h_t)}\mathbb{P}_{M_0}(h_t)$. So a δ-correct test satisfies $\mathbb{P}_{M_0}(C_\tau \le c) \le \delta(1 + e^c)$: it must collect honest cost about b before it stops.

(T3) *Sorting.* Moving a success probe before a failure probe raises the belief at which that failure is sent, so by (K1) it raises the honest cost. Among probe sequences with a given number of successes, the cost is largest for all successes first and then all failures. Comparing the resulting sum with an integral gives a cost of at most $(s + n)F(m)$ plus a logarithmic term, where m is the final belief.

(T4) *Waiting times.* Between two pulls the state of the agent is frozen, so the number of rounds until the next pull is geometric with the current pull probability. The expected number of rounds to make n pulls is $\sum_{k<n} 1/p_k$.

Sketch of the proof of Proposition 2. Without probes the test sees only the choices. Let E_T be the event that the agent pulls at every round up to T. On E_T the observed sequence is $(1, \dots, 1)$ under both models, so any test stops and decides on this sequence as a function of its own randomness only. A short *forced-path* argument then gives

$$\mathbb{P}_0(\hat{z} = 1) + \mathbb{P}_1(\hat{z} = 0) \;\geq\; \min\big(\mathbb{P}_0(E_T), \mathbb{P}_1(E_T)\big)\, \mathbb{P}(\text{the test stops before } T \text{ on } 1, 1, \dots).$$

It remains to show that both models pull forever with probability bounded away from zero. Under the unbiased agent facing an option of rate 1, every outcome is a success, the belief tends to 1 and the skip probabilities are summable by (T1). Under the biased agent, (K4) makes the perceived success probability at least the current belief above the lock-in level ℓ. A coupling with a Pólya urn, whose mean is a martingale, together with Doob's maximal inequality, shows that after a fast start the belief stays above ℓ forever with probability at least $\frac{1}{2}$. There the skip probabilities are again summable. Letting $T \to \infty$ gives an error sum at least $\rho > 0$, so no test is δ-correct for $\delta < \rho/2$. The values of ρ in the main text are computed numerically.

Sketch of the proof of Proposition 3. We prove: for a biased agent with threshold w and a known prior, against the null of unbiased agents with any threshold $w_0 \in (0, w]$, every design and test that stops under M_1 has $\sup_{w_0} \mathbb{P}_{M_{0,w_0}}(\hat{z} = 1) + \mathbb{P}_{M_1}(\hat{z} = 0) \geq p_0$, where p_0 is the limiting probability that the biased agent pulls forever when all probes are failures. Along the all-pull path the probes, and hence the whole observed history, are a function of the auditor's randomness alone, so the forced-path argument applies again. A pathwise coupling shows that success probes can only raise the biased agent's perceived count, so by (T1) it pulls forever with probability at least p_0, which is positive by the lock-in argument above. An unbiased agent with a very small threshold w_0 pulls at each of the first T rounds with probability close to 1, whatever the probes. Choosing w_0 small enough for each T and letting $T \to \infty$ gives the claim. The argument needs the auditor to have no positive lower bound on the threshold.

Sketch of the proof of Theorem 4. Let $\theta_K(q) = \mathrm{kl}(q, w)/e_K(q)$. The statement covers four parts: (i) under (K1) and (K5), every δ-correct test has $\mathbb{E}_{M_0} N_\tau \geq (1 - o(1))\, b/e_K(q)$; (ii) if moreover $q < w$, then $\log \mathbb{E}_{M_0} \tau \geq (1 - o(1))\, \theta_K(q)\, b$, and the same holds for the median; (iii) under (K2) and $q = w$, $\mathbb{E}_{M_0} \tau \geq c\, b^2$; (iv) for $q < w$, the exact sequential likelihood ratio test makes at most $(1 + o(1))\, b/e_K(q)$ pulls in expectation and stops before $\delta^{-\theta_K(q) - \varepsilon}$ rounds with probability tending to one.

(i) Probes. With a stationary design the probes are i.i.d. Bern(q), so the honest belief concentrates at q. After n pulls the honest cost is $n(1-q)\ell_0(q) = n\, e_K(q)$ up to $o(n)$, using the continuity in (K5). By (T2), a test that stops before $(1-\varepsilon)b/e_K(q)$ pulls has honest cost below $(1-\varepsilon)b$ on a likely event, which has probability at most δ^{ε} under M_0. So $N_\tau \geq (1-\varepsilon)b/e_K(q)$ with high probability.

(ii) Rounds for $q < w$. After n pulls the honest belief is near $q < w$, so by (T1) and a Chernoff bound the pull probability is at most $e^{-n\, \mathrm{kl}(q,w)(1-o(1))}$. By (T4) the wait before the next pull is geometric with this parameter. Since at least $b/e_K(q)$ pulls are needed by (i), the last waits are of order $\exp(b\, \mathrm{kl}(q, w)/e_K(q)) = \delta^{-\theta_K(q)}$.

(iii) The boundary $q = w$. Here the honest belief does not move away from w, but the test can be unlucky. Consider the probe paths whose empirical success rate after $k \leq \beta_0 b$ pulls lies in $[w/4, w/2]$. Their probability is about $e^{-k\, \mathrm{kl}(w/4, w)}$, which is much larger than δ, and their honest cost is at most $b/2$. By (T2) the test cannot stop on most of them before k pulls. On such a path the next wait has mean $1/G$, and a direct computation with the binomial form of G shows that $\mathbb{P}(X_k = x)/G(A + x, B + k - x)$ is bounded below by a constant. Summing over $k \leq \beta_0 b$ and over the order of $wk/4$ admissible values of x gives $\mathbb{E}_{M_0} \tau \geq c\, b^2$.

(iv) The exact test. The likelihood ratio after n pulls is a mixture, over the latent perceptions of the biased agent, of products of ratios of geometric waiting-time probabilities. We bound its moment of order $\eta \in (0, 1)$ under M_0. We split the latent paths at the last epoch where the biased agent has few flips (failures perceived as successes). Paths with few flips have probability at most $e^{-n\, e_K(q)(1-o(1))}$, by a count of the flip sets with the binary entropy. Epochs with many flips give the biased agent a pull probability much larger than the honest one, and each such waiting factor has a small moment. This gives $\mathbb{E}_{M_0}[L^\eta] \leq (n+1) e^{-\eta n\, e_K(q)(1-o(1))}$, and Markov's inequality bounds the number of pulls by $b/e_K(q) + O(\log b)$. The rounds follow from (T4), since the honest pull probability after k pulls is at least $e^{-k(\mathrm{kl}(q,w) + o(1))}$. Correctness follows from Ville's inequality applied to the likelihood ratio and its inverse, which are positive martingales under M_0 and M_1. Parts (i) and (ii) apply to this test, so it matches both bounds to first order.

Sketch of the proof of Theorem 5. We use (K1) to (K3) and a sharp choice rule: (S1) away from w by η, after n pulls, the agent pulls or skips with probability $1 - O(e^{-c_\eta n})$; (S2) the pull probability is nondecreasing when a failure is turned into a success. Thompson sampling satisfies both by (T1). Fix $\varepsilon > 0$ and $T = (1 - 2\varepsilon)b/F(w)$. We show $\mathbb{P}_0(\tau \le T) \to 0$, which gives $\mathbb{E}_0\tau \ge (1 - o(1))T$.

Step 1. By (S1), under M_0, with probability $1 - O(1/T)$ the honest agent never pulls from a mean below $w - \eta$ and never skips from a mean above $w + \eta$ once it has $O(\log b)$ pulls.

Step 2. Let σ be the last pull at which the honest mean is at most $w + \eta$. After σ the agent pulls at every round. A comparison of the two models then gives $p_1(h) \ge e^{-C_\sigma} p_0(h)$: keep the term of M_1 in which the first σ probes are perceived correctly, and bound the later pull factors of M_1 below by those of M_0 using (S2), since the biased agent has at least as many perceived successes.

Step 3. At pull σ the mean is within 2η of w. By the sorting bound (T3), the honest cost of the first $\sigma \le T$ pulls is at most $T\,F(w)(1 + O(\eta)) + O(\log b \log\log b) \le (1 - \varepsilon)b$.

Step 4. By the change of measure, $\mathbb{P}_0(\tau \le T) \le \delta + \delta^\varepsilon + O(1/T) \to 0$. Let $\eta \to 0$ and then $\varepsilon \to 0$, using the continuity of F at w.

Sketch of the proof of Theorem 6. The fill-drain design $\mathrm{FD}(r)$ sends success probes at the first r pulls and failure probes afterwards. With $R = A + r$, the honest mean after $j \ge r$ pulls is $R/(s + j)$, and it crosses w at the pull $n_c(r)$. The window test waits for pull n_c, watches the next W rounds, and declares the agent biased if it sees at least M pulls.

Drain information. The honest cost of the drain, $I(r) = \sum_{j=r+1}^{n_c} \ell_0(R/(s + j - 1))$, is a Riemann sum of the integral that defines F. So $I(r) = n_c F(w) + O(\log n_c)$. We choose the smallest r with $I(r) \ge b + O(b^{2/3}\log b)$, which gives $n_c = b/F(w) + O(b^{2/3}\log b)$. We take $M = L$ and $W = L + V$ with L, V of order $b^{2/3}$.

No false alarm. Under M_0, after k window pulls the honest mean is below w by about wk/n_c. By (T1) the pull probability is at most $\exp(-w^2k^2/(8(s + n_c)))$. Seeing L pulls therefore has probability at most $W^L \exp(-cL^3/b)$, which is at most δ for L of order $b^{2/3}$.

Power. Under M_1 the number D of drain failures perceived as successes dominates a sum of independent Bernoulli variables with success probabilities $g(\mu_{j-1})$. So $\mathbb{P}_1(D < d) \le d\,\mathrm{poly}(b)^d\, e^{-I(r)} \le \delta/2$ for d of order $b^{2/3}$. When $D \ge d$, the latent mean during the window stays at least $w + \frac{1}{2}b^{-1/3}$. By (T1) each skip then has probability at most $\exp(-c\,b^{1/3})$, and seeing $V + 1$ skips has probability at most $\delta/2$.

Cost. The test stops at round $T_{n_c} + W$. The skips before pull n_c add $O(\sqrt{n_c})$ rounds in expectation by (T4). This step uses the following fact, which we do not prove: *Fact P*, there are $c_0 > 0$ and N_0 with $G(\alpha, \beta) \ge c_0$ whenever $\alpha/(\alpha + \beta) \ge w$ and $\alpha + \beta \ge N_0$. Together, $\mathbb{E}_z\tau = b/F(w) + O(b^{2/3}\log b)$ under both models, which matches Theorem 5. The upper bound on $\mathbb{E}_z\tau$ rests on Fact P.

Deterministic learners. The last sentence of Theorem 6 concerns learners with $\pi \in \{0, 1\}$, where $\pi(\alpha, \beta)$ is the pull probability. We use the following conditions on the rule π. (D0) $\pi(A, B) = 1$. (D1) If $\pi(\alpha, \beta) = 1$ then $\pi(\alpha + 1, \beta) = 1$. Together with (S2) they hold for greedy choice $\mathbf{1}\{m > w\}$ when $A/s > w$, and for $\mathbf{1}\{m + \mathrm{bonus}(n) > w\}$ when the bonus is nonincreasing in n and $n \cdot \mathrm{bonus}(n)$ is nondecreasing. The state changes only at pulls, so after the first skip the agent skips forever. Call a design and a test deterministic if $q_t \in \{0, 1\}$ depends only on the past actions and the test uses no randomisation. For a probe sequence ξ and k pulls, let $I_k(\xi) = \sum_{j \le k,\ \xi_j = 0} \ell_0(\mu_{j-1})$.

Proposition 9 (Greedy and count-based learners)**.** *Assume $K(1, \cdot) \equiv 1$, (K1), (K2), (D0), (D1), (S2) and $\delta < \frac{1}{2}$. Let $n_c(r)$ be the honest freeze count of* $\mathrm{FD}(r)$*, and $I(r) = I_{n_c(r)}$ for its probe sequence.*

(i) Under $\mathrm{FD}(r)$ *with* $\mathrm{WIN}(W, M)$ *for any $W, M \ge 1$, $\mathbb{P}_0(\hat{z} = 1) = 0$ and $\tau = n_c(r) + W$ under both models.*

(ii) With $W = M = 1$, $\mathbb{P}_1(\hat{z} = 0) = e^{-I(r)}$.

(iii) Every deterministic design with a deterministic δ-correct test has $\tau \ge k^* + 1$, where $k^* = \min\{k \colon \bar{I}(k) \ge b\}$ and $\bar{I}(k)$ is the largest value of I_k over sequences that freeze M_0 exactly at k. The design $\mathrm{FD}(r)$ *with the smallest r such that $I(r) \ge b$, with $W = M = 1$, is δ-correct and attains $\tau = k^* + 1$.*

(iv) For greedy choice, $k^* = b/F(w) + O(\log b)$.

Sketch of the proof. With a deterministic design the honest history is one fixed sequence: the agent pulls k_0 times and then freezes. By (S2) the biased agent pulls whenever the honest one does. If it perceives none of the k_0 failures as a success, an event of probability $e^{-I_{k_0}(\xi)}$, its history is identical to the honest one, and the test must say "unbiased" there. If it perceives at least one, (D1) and (S2) show that it pulls at round k_0+1, which the honest agent never does. This gives (i) and (ii), and shows that a δ-correct test needs $I_{k_0}(\xi) \geq b$. By the sorting argument (T3), which keeps the freeze count unchanged, I_k is largest for the fill-drain sequence, which gives (iii). Part (iv) is the drain-information estimate of the previous sketch.

Sketch of the proof of Theorem 7. The prior is only known to lie in a box $[A_{\mathrm{lo}}, A_{\mathrm{hi}}] \times [B_{\mathrm{lo}}, B_{\mathrm{hi}}]$, and the biased kernel is only known to satisfy $K(0,\cdot) \geq g_{\min}$. Let $\pi^+ = (A_{\mathrm{hi}}, B_{\mathrm{lo}})$ and $\pi^- = (A_{\mathrm{lo}}, B_{\mathrm{hi}})$ be the optimistic and pessimistic corners, and $\Delta = (A_{\mathrm{hi}} - A_{\mathrm{lo}})\frac{1-w}{w} + (B_{\mathrm{hi}} - B_{\mathrm{lo}})$.

Lower bound. The floor kernel $g_{\min}$ is admissible. A uniformly δ-correct audit is in particular δ-correct for the pair with any fixed prior and kernel $g_{\min}$, so Theorem 5 gives $\mathbb{E}\tau \geq (1-o(1))\, b/F_{\min}(w)$.

Corner reduction. We represent the process by pulls: independent uniforms drive the perceptions and the geometric waits, and the window count is nonincreasing in every wait. Coupling two models through the same uniforms shows that a larger pull probability at every pull gives a larger window count. By (T1) and (K1), the false-alarm probability over the box is largest for the honest agent with prior π^+, and the missed-detection probability is largest for the biased agent with prior π^- and kernel $g_{\min}$. The expected cost is at most $\sum_{k<k_a} 1/p_k^- + W$, where p_k^- is the honest pull probability at the pessimistic corner and k_a is the pull at which the window starts.

Upper bound. We use FD(r) with the window anchored at the pessimistic crossing n^- and the threshold $M = (n^+ - n^-) + L$, which covers the optimistic crossing n^+. The two crossings differ by at most $\lceil \Delta \rceil$. The two corner errors are then bounded as in the previous sketch, with $g_{\min}$ and $F_{\min}$ in place of g and F. For the cost, Fact P is replaced by an anti-concentration bound that we prove: $G(\alpha,\beta) \geq c_w > 0$ whenever $\alpha \geq 2$ and $\alpha/(\alpha+\beta) \geq w$, from the binomial form of G and the central limit theorem. This gives

$$\mathbb{E}\tau \leq \frac{b}{F_{\min}(w)} + O(b^{2/3}\log b) + \lceil \Delta \rceil\, O(\log b),$$

which matches the lower bound. The price of the unknown prior is the term in $\lceil \Delta \rceil$, which does not grow like b.

Exact calibration at every δ. The constants of the upper bound are asymptotic. Numerical checks with the explicit constants found them feasible only from b of the order of $1e+06$ for the strong kernel on the box $[1.5,3]^2$. The corner reduction does not need them. Let (r, k_a, W, M) be any parameters with $k_a > r$ such that the two corner errors are at most δ. These errors are computed exactly by dynamic programming, as a pure-birth chain under M_0 and a recursion on (window pulls, latent count) under M_1. Then the corner reduction shows that the test is uniformly δ-correct over the box and all kernels above $g_{\min}$, with expected cost at most $\sum_{k<k_a} 1/p_k^- + W$, for every δ. The search over parameters uses only the inputs $(w, \mathrm{box}, g_{\min}, \delta)$, so it introduces no selection effect. We do not claim that the result is optimal within the family.

Sketch of the proof of Proposition 8. An honest agent with false-positive rate v reads a failure probe as a success with probability v, independently of its belief. Drive all values of v with the same uniforms. For $v \leq v'$ the agent with rate v' has at least as many perceived successes at every pull, so by (T1) it has a larger pull probability at every pull, shorter waits and a larger window count. Hence the false-alarm probability is nondecreasing in v, and calibrating at $v_{\max}$ covers every $v \leq v_{\max}$. Combined with the corner reduction, calibrating at $(\pi^+, v_{\max})$ covers every prior in the box. The cost bound is unchanged. The argument uses that a success probe is always read as a success. A grader that also reads successes as failures is not covered.

A.3 Finite-δ bounds

The first-order rate describes the limit of small δ. At practical values of δ it understates the cost by a factor of four to eight. The following bounds hold for every $b > 0$ and are computed exactly.

Proposition 10 (Finite-δ sandwich)**.** *For the strong kernel and $w = \frac{1}{2}$, every δ-correct test and design satisfies $\mathbb{E}_{M_0}\tau \geq L(b)$ for every $b > 0$, where $L(b)$ is the value of a Markov decision process over adaptive designs in which information is counted only up to the last skip. For $b \geq 2$ the calibrated fill-drain audit satisfies $\mathbb{E}_{M_z}\tau \leq U(b)$, where $U(b)$ comes from the proof of Theorem 7 with exact binomial tails.*

Table 2 lists the values for a known prior $(2, 2)$. The audit lies between the bounds at every point we computed.

$\log(1/\delta)$	lower	audit	upper
2.0	2.1	9.5	15.0
7.6	9.3	21.2	44.4
28.7	27.7	52.5	107.2
108.7	88.9	145.5	273.5

Table 2: Lower bound, calibrated audit, and upper bound on the expected rounds for the strong kernel and a known prior $(2, 2)$.

B EXPERIMENT DETAILS

B.1 Simulation settings

Agents are Thompson sampling learners with a Beta prior, $w = \frac{1}{2}$, and the strong $(.5, 1, 2)$, mild $(.3, 20, 2)$, and weak $(.1, 5, 2)$ kernels. Random-canary runs use 100 paths per setting and a fixed cap on rounds and probes, and unfinished runs are counted as censored. The sweep of fixed pass rates in Table 1(a) runs each rate with 100 paths split into four batches. The noise table uses the exact calibration with the null set of honest graders with $v \le v_{\max}$ and no sampling. The stress test uses 20000 runs per agent type at $\delta = 10^{-2}$ and the prior box $[1.5, 3]^2$. The simulator was checked against the exact error probabilities of the calibrated design. At the optimistic corner it gives a false-alarm rate of 0.0067 against 0.0069 exactly, and at the pessimistic corner a miss rate of 0.0069 against 0.0070 for the strong kernel.

B.2 Price of an unknown prior

Figure 4 compares the audit calibrated over the prior box with an auditor that knows the worst prior, for three kernels, two boxes, and $\delta \in \{10^{-1}, \dots, 10^{-4}\}$. The gap is a constant number of rounds. The ratio of the audit cost to $b/F_{\min}(w)$ approaches one slowly as b grows. At $b = 128$ it is 1.81 for a known prior and 1.95 for the box. The closed-form upper bound of Theorem 7 can only be evaluated for $b \ge 1e + 06$.

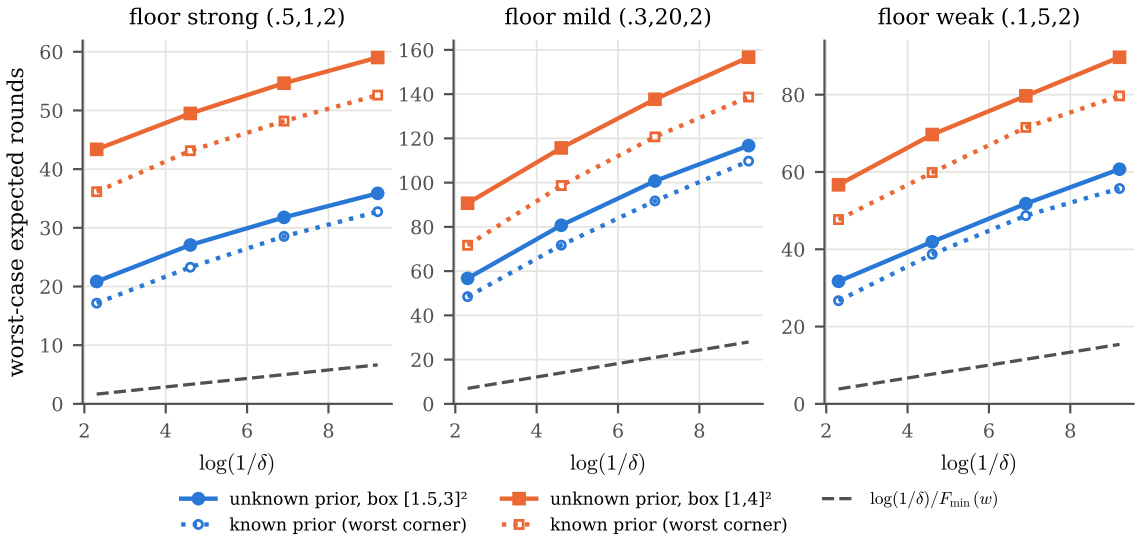


Figure 4: Rounds of the audit calibrated over the prior box and of the audit with a known worst prior, against δ.

B.3 Wrong default value

Figure 5 shows the larger of the false-alarm and miss rates when the true threshold w' of the agent differs from the value $w = \frac{1}{2}$ used by the audit, for δ from 10^{-1} to 10^{-4}. Each point uses 100000 runs per agent type and the box audit of Section 7. Misses stay close to δ when $w' > w$. False alarms exceed δ when $w' < w$, and they do so at smaller offsets for the mild kernel. A safety margin on the window removes them, as in the stress test. This is the graded version of Proposition 3, which says that no audit works when w is unknown.

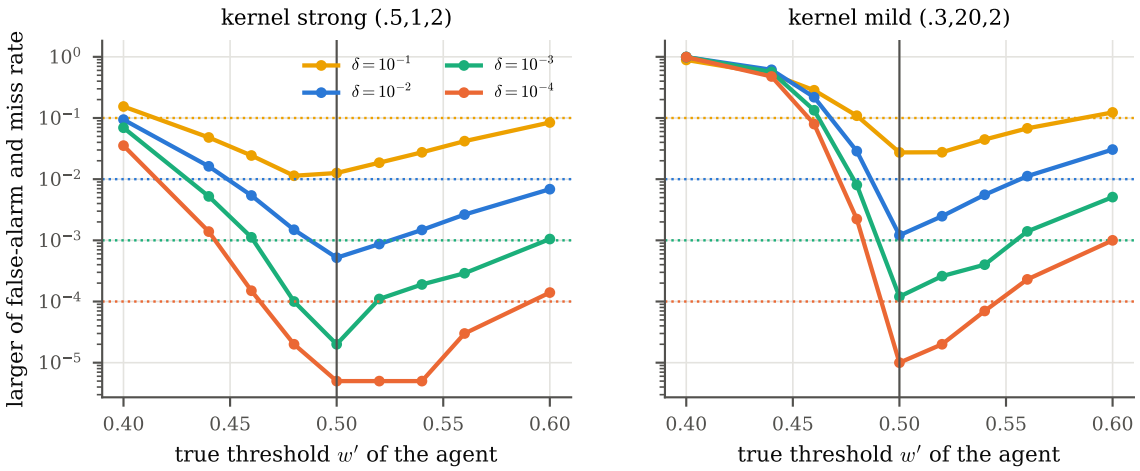


Figure 5: Larger of the false-alarm and miss rates against the true threshold w' of the agent, for the audit built for $w = \frac{1}{2}$. Dotted lines mark δ.